



# Dyslipidemias

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CPhA strives to provide guidance for health-care professionals on the appropriate care of all individuals. However, we recognize that in this chapter, our content is based on clinical research studies of patients who may not reflect the identity of the individual patient presenting. Other resources exist that may be helpful, such as the World Professional Association for Transgender Health’s (WPATH) Standard of Care. When the terms “women,” “men,” “female” and “male” are used in the text, they are used to maintain accuracy with what is reported in the clinical literature. We are aware these terms are not inclusive of all patients.

## Introduction

Dyslipidemias are abnormalities in the quantity or function of lipids or lipoproteins in the blood. The majority of dyslipidemias result from interaction between environmental and genetic factors. Understanding the contribution of each factor in the dyslipidemic patient is beyond the scope of this chapter. Nevertheless, it is essential to rule out secondary causes of dyslipidemia that, when identified and potentially eliminated, can prevent unnecessary treatment. In addition, the use of lipid-lowering agents in the management of cardiovascular (CV) disease has become part of routine preventive care.

Table 1: Factors Associated with Increased Cardiovascular Risk

|                                                                                                |                                                     |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Abdominal aortic aneurysm                                                                      | Family history of dyslipidemia                      |
| Chronic kidney disease (eGFR <60 mL/min/1.73 m <sup>2</sup> )                                  | Family history of premature coronary artery disease |
| Chronic obstructive pulmonary disease                                                          | HIV infection                                       |
| Clinical manifestations of hyperlipidemias (xanthomas, xanthelasma, premature arcus cornealis) | Hypertension                                        |

|                             |                                                                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Current cigarette smoking   | Hypertensive disorders of pregnancy <sup>[a]</sup>                                                                                                  |
| Diabetes mellitus           | Inflammatory diseases (ankylosing spondylitis, inflammatory bowel disease, psoriatic arthritis, rheumatoid arthritis, systemic lupus erythematosus) |
| Erectile dysfunction        | Obesity (BMI >30 kg/m <sup>2</sup> )                                                                                                                |
| Evidence of atherosclerosis |                                                                                                                                                     |

<sup>[a]</sup> Screen in the late postpartum period.

Screening of the lipid profile is recommended in children with a family history of dyslipidemia or CV disease, men ≥40 years of age and women ≥40 years of age or postmenopausal (see [Criteria for Intervention](#)). At-risk individuals, such as those who are South Asian, First Nations or recent immigrants, may be screened at an earlier age.<sup>[1]</sup> Current CV risk calculators have not been validated in these patients.<sup>[2]</sup> While patients who are South Asian,<sup>[3]</sup> First Nations<sup>[4]</sup> and new immigrants are considered at-risk for CV disease, the risk should not be attributed solely to biological or genetic factors. Rather, structural inequalities are known to limit racialized patients’ access to social determinants of health and contribute to disease risk.

Screen all patients with the risk factors in [Table 1](#) irrespective of age.<sup>[1]</sup>

Patients with severe hypercholesterolemia (LDL-C ≥95<sup>th</sup> percentile for age and sex or ≥5 mmol/L in adults), severe hypertriglyceridemia (HTG) (triglycerides [TG] ≥10 mmol/L), or a history of HTG-induced acute pancreatitis should be referred to a specialized lipid clinic or a genetic dyslipidemias clinic in order to receive proper diagnosis and follow-up.

## Goals of Therapy

- Reduce risk of cardiovascular diseases
- Prevent pancreatitis from severe hypertriglyceridemia

## Investigations

Clinical evaluation should focus on:

- Medical history:

- past or present CV disease, e.g., angina, aortic aneurysm, claudication, erectile dysfunction, MI, revascularization, stroke or transient ischemic attack (TIA)
- possible causes of secondary dyslipidemia (see [Table 2](#))
- major CV risk factors, e.g., diabetes mellitus, hypertension, smoking
- history of severe abdominal pain or acute pancreatitis
- Family history:
  - premature CVD (<55 y in males, <65 y in females) in first-degree relatives
  - dyslipidemia
- Physical examination:
  - waist circumference
  - bilateral brachial BP
  - fundoscopy (lipemia retinalis, retinopathies)
  - cardiovascular evaluation (bruits, peripheral pulses)
  - hepatosplenomegaly
  - lipid deposits (premature arcus cornealis, xanthomas, xanthelasmas)
- Laboratory tests:
  - lipid and lipoprotein levels, including total cholesterol, TG, high-density lipoprotein cholesterol (HDL-C) and Apolipoprotein B (ApoB) (optional) measurement, and calculated low-density lipoprotein cholesterol (LDL-C) and non-HDL-C

Note that the Friedewald equation  $\text{LDL-C} = \text{total-C} - (\text{HDL-C} + \text{TG} / 2.2)$  cannot be used if triglyceride levels are >4.52 mmol/L, if type III dysbetalipoproteinemia or if chylomicrons are present.

  - use the same laboratory for repeated measurements
  - fasting is no longer required for lipid and lipoprotein measurements unless triglycerides are >4.5 mmol/L
  - obtain 2 or 3 measurements at 4- to 6-week intervals to establish a baseline before initial diagnosis of a dyslipidemic phenotype
  - measurement of lipoprotein (a) level should be done once in a person's lifetime as a part of the initial lipid screening
  - for patients with triglycerides >1.5 mmol/L, non-HDL-C or ApoB should be used instead of LDL-C as the preferred lipid parameter for screening.<sup>[1]</sup> These particular tests are associated with atherogenic potential and have shown a stronger correlation with CV events than LDL-C. While apolipoprotein B testing may not be widely available, non-HDL-C can be calculated from routine tests (non-HDL-C = total cholesterol – HDL-C)
  - fasting blood glucose or glycated hemoglobin (HbA<sub>1c</sub>)
  - estimated glomerular filtration rate (eGFR)
  - urine albumin/creatinine ratio (optional in diabetes, hypertension or if eGFR <60 mL/min/1.73 m<sup>2</sup>)
  - other lab investigations to rule out frequent causes of secondary dyslipidemias (see [Table 2](#)) and establish baseline levels for liver transaminases and CK

**Table 2:** Common Causes of Secondary Hyperlipidemia

| Condition                        | Medication <sup>[a]</sup>                                                  |
|----------------------------------|----------------------------------------------------------------------------|
| Alcohol excess (>1–2 drinks/day) | Beta-blockers without intrinsic sympathomimetic or alpha-blocking activity |
| Chronic renal failure            | Corticosteroids                                                            |
| Diabetes/metabolic syndrome      | Highly active antiretroviral therapy (HAART)                               |
| Hypothyroidism                   | Hormone replacement therapy (HRT)                                          |
| Nephrotic syndrome               | Oral contraceptives                                                        |
| Obstructive liver disease        | Thiazide diuretics                                                         |
| Overweight/obesity               |                                                                            |
| Pregnancy                        |                                                                            |

<sup>[a]</sup> Medication-induced dyslipidemia should not preclude the use of these medications if clinically indicated.

Criteria for Intervention

In its 2021 guidelines, the Canadian Cardiovascular Society introduced the concept of using treatment thresholds for lipid-lowering therapy intensification rather than the treat-to-target approach. It is recommended that patients with high cardiovascular risk be considered to have a statin-indicated condition (see [Table 3](#)). Other patients should be stratified into 1 of 3 categories of CV risk using the Framingham Risk Score (FRS). It is recommended to double the FRS when there is a family history of premature CV disease. This is referred to as the modified FRS (see [Figure 1](#)). Based on the risk score, thresholds to initiate and intensify treatment are shown in [Figure 2](#).<sup>[1]</sup> Reassess the cardiovascular risk using the modified FRS every 5 years for patients who are 40–75 years of age.

Coronary artery calcium (CAC) screening using computed tomography may be considered for asymptomatic adults ≥40 years of age with an intermediate FRS and for whom treatment decisions are uncertain (see [Figure 2](#)).<sup>[1]</sup> Consider referring these individuals for specialist assessment.

High triglyceride levels (2.3–11.3 mmol/L) are an additional CV risk factor when associated with atherogenic dyslipidemias, e.g., familial combined hyperlipidemia, insulin resistance, diabetes, chronic kidney disease. Very high triglyceride levels (>10 mmol/L) are a risk factor for pancreatitis.

**Table 3:** High Cardiovascular Risk Conditions Where Treatment with a Statin is Recommended<sup>[1]</sup>

| Condition                              | Description                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atherosclerotic cardiovascular disease | Myocardial infarction, acute coronary syndromes, stable angina, documented coronary disease by angiography (>10% stenosis), stroke, TIA, documented carotid disease, peripheral artery disease, claudication and/or ABI <0.9, |

| Condition                                                                                                     | Description                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                               | abdominal aortic aneurysm (abdominal aorta >3 cm <i>or</i> previous aneurysm surgery)                                                                                                           |
| Chronic kidney disease                                                                                        | >3 months duration and ACR >3 mg/mmol L <i>or</i> eGFR <60 mL/min/1.73m <sup>2</sup> and ≥50 years of age                                                                                       |
| Diabetes mellitus                                                                                             | ≥40 years of age <i>or</i> >15 years' duration and ≥30 years of age <i>or</i> microvascular complications                                                                                       |
| LDL-C ≥5 mmol/L                                                                                               | LDL-C ≥5 mmol/L <i>or</i> ApoB ≥1.45 g/L <i>or</i> non-HDL-C ≥5.8 mmol/L <i>or</i> documented familial hypercholesterolemia (FH) <i>or</i> other genetic dyslipidemia; exclude secondary causes |
| Abbreviations: ABI = ankle brachial index; ACR = albumin-to-creatinine ratio; TIA = transient ischemic attack |                                                                                                                                                                                                 |

## Therapeutic Choices

### Nonpharmacologic Choices

- Lipid-lowering therapy is a multifaceted treatment that requires patient engagement. Efforts should be made to have the patient participate in decision-making. Using the concept of “cardiovascular age” in discussions can help to encourage this.<sup>[1]</sup> An online calculator of cardiovascular age is available at [myhealthcheckup.com](https://myhealthcheckup.com).
- Diet aimed at reducing blood lipid levels and weight (if needed) should always be part of the treatment. For secondary prevention and high-risk individuals, cholesterol-lowering medication is introduced simultaneously. For primary prevention, a 3-month dietary trial (see [Table 4](#)) is recommended before considering medication; during this time, take 2 (ideally 3) lipid and lipoprotein measurements. Involving a registered dietitian in the initial evaluation as well as the follow-up of the patient is strongly recommended.
- Other lifestyle changes are important to further modify the lipoprotein profile and reduce the risk of CVD (see [Table 4](#)).
- Where available, LDL apheresis or plasmapheresis may be used in specialized lipid clinics to lower LDL-C in patients with homozygous familial hypercholesterolemia. This involves the separation and removal of blood lipids using filtration or centrifugation. Plasmapheresis or plasma exchange may be used to lower triglycerides in patients with very severe hypertriglyceridemia.

**Table 4:** Dietary and Lifestyle Interventions Associated with an Improved Lipid Profile or Reduced Cardiovascular Disease Risk<sup>[1]</sup>

| Food-Based Interventions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Nutrient-Based Interventions <sup>[a]</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Other Lifestyle Interventions                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dietary patterns: <ul style="list-style-type: none"><li>• DASH<sup>[b]</sup></li><li>• Low-glycemic index/glycemic load<sup>[b]</sup></li><li>• Mediterranean</li><li>• Portfolio<sup>[b]</sup></li><li>• Vegetarian</li></ul> Encourage the consumption of: <ul style="list-style-type: none"><li>• Nuts (≥30 g/day)<sup>[b]</sup></li><li>• Legumes (≥4 servings/wk to ≥1 serving/day)<sup>[b]</sup></li><li>• Olive oil (≥60 mL/day)</li><li>• Fruits and vegetables (≥5 servings/day)</li><li>• Whole grains (≥3 servings/day)</li><li>• Soy protein (≥30g/day)<sup>[b]</sup></li></ul>       | Reduce the intake of: <ul style="list-style-type: none"><li>• Saturated fats<sup>[b]</sup><ul style="list-style-type: none"><li>◦ NCEP Step I: ≤10% energy</li><li>◦ NCEP Step II: ≤7% energy</li></ul></li><li>• Dietary cholesterol<sup>[b]</sup><ul style="list-style-type: none"><li>◦ NCEP Step I: ≤300 mg/day</li><li>◦ NCEP Step II: ≤200 mg/day</li></ul></li><li>• Alcohol (1–2 drinks/day)</li></ul> Increase the proportion of mono- and polyunsaturated fats<br><br>Increase the intake of: <ul style="list-style-type: none"><li>• Total fibre (≥30 g/day)</li><li>• Viscous soluble fibre (≥10 g/day)<sup>[b]</sup></li><li>• Omega-3 (EPA and DHA [2–4 g/day]) (to lower TG only)</li></ul> Supplementation of plant sterols/stanols (≥2 g/day) <sup>[b][c]</sup> | Weight loss and reduction of abdominal obesity (5–10% of body weight loss) <sup>[b]</sup><br><br>Physical activity (30–60 min/day moderate to vigorous intensity)<br><br>Smoking cessation<br><br>Sufficient duration of sleep (6–8 h per night)<br><br>Stress management |
| <sup>[a]</sup> This table does not include the dietary interventions recommended for patients with severe hypertriglyceridemia (chylomicronemia), for which the main goal is to reduce the risk of acute pancreatitis.<br><sup>[b]</sup> Interventions recommended to lower LDL-C.<br><sup>[c]</sup> Natural sources are unable to provide a high enough intake, so specific supplements (e.g., Cardiosmile) or sterol-enriched foods must be used, e.g., margarine, fruit juice.<br>Abbreviations: DASH = dietary approaches to stop hypertension; NCEP = national cholesterol education program |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                           |

### Pharmacologic Choices

[Table 5](#) summarizes the effects of lipid-lowering drugs on lipoproteins. Statins are recommended as first-line lipid-lowering agents for cardiovascular event prevention.<sup>[1]</sup> Other agents from different classes may be substituted if statins are not tolerated or added if threshold lipid targets are exceeded. However, cardiovascular outcomes are not clearly improved with all nonstatin agents. Some combination therapies (such as fibrates and statins) carry an increased risk of drug toxicity, particularly with long-term use, so consider referral of such patients to a specialized lipid clinic.<sup>[9]</sup>

**Table 5:** Lipid-Lowering Agents’ Effect on Lipoproteins<sup>[1][6][7][8]</sup>

| Lipid-Lowering Agent                   | LDL-C    | HDL-C    | Triglycerides          |
|----------------------------------------|----------|----------|------------------------|
| HMG-CoA Reductase Inhibitors (Statins) | ↓ 20–60% | ↑ 5–15%  | ↓ 7–30% <sup>[a]</sup> |
| Ezetimibe                              | ↓ 14–25% | ↑ 1%     | ↓ 7–9%                 |
| Fibrates                               | ↓ 5–20%  | ↑ 10–35% | ↓ 20–50%               |
| PCSK9 Inhibitors                       | ↓ 50–70% | ↑ 5–10%  | ↓ 15%                  |
| Resins                                 | ↓ 10–30% | ↑ 3–10%  | ↑ 10–25%               |

<sup>[a]</sup> Atorvastatin and rosuvastatin have the greatest triglyceride-lowering effect.

See [Table 7](#) for prescribing information on drugs used in the management of dyslipidemias.

**HMG-CoA Reductase Inhibitors (Statins)**

HMG-CoA reductase inhibitors are potent LDL-lowering agents that interfere with the atherosclerotic disease process. Effect on HDL-C is modest. Significant reductions in CVD morbidity, CVD mortality and total deaths have been associated with their use. Over 1 year, for every 1 mmol/L reduction in LDL-C, the relative risk of major vascular events is reduced by 21% and the relative risk of all-cause mortality is reduced by 10%.<sup>[10]</sup> In another analysis, this relative risk reduction was lower in the first year of therapy (about 9%) and increased to 22–25% for each subsequent year of statin therapy.<sup>[11]</sup> For primary CVD prevention, statins should be initiated at the recommended starting doses to ensure they are tolerated and titrated to reach target doses. For secondary CVD prevention, the maximal dose of a high-intensity statin should be used. Statin therapy can be divided according to lipid-lowering activity:

- Low-intensity therapy (LDL-C reduction <30%) includes fluvastatin 20 and 40 mg, lovastatin 20 mg, pravastatin 10 and 20 mg, and simvastatin 5 and 10 mg.
- Moderate-intensity therapy (LDL-C reduction 30–50%) includes atorvastatin 10 and 20 mg, fluvastatin 80 mg, lovastatin 40 and 80 mg, pravastatin 40 and 80 mg, rosuvastatin 5 and 10 mg, and simvastatin 20 and 40 mg.
- High-intensity therapy (LDL-C reduction ≥50%) includes atorvastatin 40 and 80 mg, and rosuvastatin 20 and 40 mg.

In secondary prevention of CVD, all patients should receive statin therapy. For primary prevention of CVD in patients at high risk of coronary artery disease, statins are also recommended.<sup>[1]</sup> Because achieving very low LDL-C levels (<1.30 mmol/L) is safe and yields additional cardiovascular benefits,<sup>[12][13]</sup> the use of high-intensity statin is preferred, particularly in secondary prevention patients.<sup>[1]</sup> Statins may be considered in patients in lower-risk groups based on the individual's global cardiovascular risk (see [Figure 1](#), [Figure 2](#)).<sup>[1][14]</sup>

Muscle-related symptoms are often reported with statins, but must be carefully assessed to rule out other causes.<sup>[15]</sup> Many factors may predispose patients to myotoxic effects (see [Table 6](#)), so close monitoring and perhaps lower doses or alternate-day dosing of statins will be required in these situations. [Figure 3](#) provides suggestions for managing muscle-related symptoms. See [Suggested Readings](#) for references reviewing this topic. A statin intolerance calculator is available at [www.circl.ubc.ca/english/web\\_statin.html](http://www.circl.ubc.ca/english/web_statin.html).

Other concerns are the risk of new-onset diabetes, estimated as 1 extra case of diabetes for each 1000 patients/years of treatment,<sup>[16]</sup> and anecdotal reports of cognitive side effects (such as memory loss). Although the potential increased risk of diabetes appears to be related to high doses of statins, the benefits of statin use exceeds the risk of diabetes in high cardiovascular risk patients. In 2014, the Statin Cognitive Safety Task Force concluded, based on low- to moderate-quality evidence, that statins as a class are not associated with adverse effects on cognition.<sup>[19]</sup>

For children with heterozygous and homozygous familial hypercholesterolemia (FH), statin therapy is recommended; the usual age for initiation is 8–10 years. The management of these patients is complex. The LDL-C threshold required to initiate statin therapy as well as the age for statin initiation require clinical judgment and should take into consideration the preferences of the patient and family.<sup>[20]</sup> Children with FH should be referred to a specialized lipid clinic.

**Table 6:** Risk Factors That May Predispose to Statin-Associated Adverse Effects<sup>[16]</sup>

| Patient Factors                        | Comorbidities                                                | Other Drugs <sup>[a]</sup>          | Statin Properties                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advanced age (>80 y)                   | Diabetes mellitus                                            | Amiodarone                          | High statin dose                                                                                                                                                                                                     |
| Alcohol abuse                          | Hepatic dysfunction                                          | Amphetamines                        | Potential for drug-drug interactions metabolized by cytochrome P450 pathways, especially 3A4 and 2C9 (likely with atorvastatin, lovastatin and simvastatin; unlikely with fluvastatin, pravastatin and rosuvastatin) |
| Asian race <sup>[b]</sup>              | History of creatine kinase elevation                         | Antipsychotics                      |                                                                                                                                                                                                                      |
| Excessive grapefruit juice intake      | History of pre-existing/unexplained muscle/joint/tendon pain | Azole antifungals                   |                                                                                                                                                                                                                      |
| Family history of myopathy             |                                                              | Cocaine                             |                                                                                                                                                                                                                      |
| Sex (women are more affected than men) |                                                              | Cyclosporine                        |                                                                                                                                                                                                                      |
| Low body mass index                    | Hypertension/heart failure                                   | Fibrates (particularly gemfibrozil) |                                                                                                                                                                                                                      |
| Nocebo effect                          | Hypothyroidism (untreated)                                   | Macrolide antibiotics               |                                                                                                                                                                                                                      |
| Perioperative periods                  | Neuromuscular diseases                                       | Nefazodone                          |                                                                                                                                                                                                                      |
| Specific genetic polymorphisms         | Severe renal disease                                         | Nicotinic acid                      |                                                                                                                                                                                                                      |
|                                        |                                                              | Protease inhibitors                 |                                                                                                                                                                                                                      |
| Vigorous exercise                      |                                                              | Unregulated supplements             |                                                                                                                                                                                                                      |
| Verapamil                              |                                                              |                                     |                                                                                                                                                                                                                      |
| Warfarin                               |                                                              |                                     |                                                                                                                                                                                                                      |

[a] This list provides significant examples of interacting drugs, but is not inclusive.

[b] According to pharmacokinetic studies, Asian individuals have higher circulating concentrations of some statins or their active metabolites than Caucasians. The mechanisms explaining this difference are not precisely understood, but may be related to differences in the prevalence of variants coding for statin-metabolizing enzymes and membrane transporters.<sup>[17]</sup>

### Cholesterol Absorption Inhibitors

**Ezetimibe**, which inhibits intestinal cholesterol absorption, is indicated as monotherapy or in combination with statins. It is better tolerated than resins, has low potential for drug interactions with cytochrome P450 substrates and does not affect the absorption of fat-soluble vitamins.<sup>[21]</sup> Early clinical trials with ezetimibe did not show improvement in outcomes such as reduced subclinical atherosclerosis or cardiovascular events,<sup>[22][23]</sup> but subsequent trials in specific scenarios have shown otherwise. In patients with chronic kidney disease, the combination of **simvastatin** and ezetimibe significantly and safely reduced the risk of major atherosclerotic cardiovascular events by 17% compared with placebo; it should be considered in this patient population.<sup>[24]</sup> In the IMPROVE-IT trial, ezetimibe in combination with simvastatin reduced the incidence of major coronary events, nonfatal stroke and death from cardiovascular causes by 6% compared to simvastatin alone in patients with

recent acute coronary syndrome.<sup>[25]</sup> Therefore ezetimibe could be considered primarily in addition to statin therapy, but also as an alternative regimen if statins are not tolerated.<sup>[1]</sup>

### Fibrates

**Bezafibrate**, **fenofibrate** and **gemfibrozil** lower triglyceride levels and raise HDL-C. The effect of fibrates on LDL-C is variable; bezafibrate and fenofibrate lower LDL-C more consistently. Except for gemfibrozil, combining a fibrate with a statin may be considered. In the FIELD trial, fenofibrate reduced nonfatal MI by 24% and coronary revascularization by 21%, but was associated with a nonsignificant increase in fatal events compared to placebo in patients with type 2 diabetes.<sup>[26]</sup> The combination of a fibrate and a statin in the type 2 diabetes population is not routinely recommended, although it could be beneficial for patients with a high baseline triglyceride level and low HDL-C.<sup>[27]</sup> In patients with severe hypertriglyceridemia (TG ≥10 mmol/L), the risk of acute pancreatitis is very high and preventive actions are mandatory. Fibrates are recommended as the first-line pharmacological therapy in these patients.<sup>[28]</sup> Combination therapy of a statin and a fibrate is often needed in patients with dysbetalipoproteinemia.<sup>[28]</sup> Referral to a specialized lipid clinic must be considered for these patients.

### PCSK9 Inhibitors

**Alirocumab** and **evolocumab** represent a class of lipid-lowering drugs called **PCSK9 inhibitors**. These agents are monoclonal antibodies directed against proprotein convertase subtilisin/kexin type 9 (PCSK9), a key protein in the LDL receptor degradation pathway. They are supplied as single-dose prefilled autoinjectors, which are administered subcutaneously either every 2 weeks or once monthly. The PCSK9 inhibitors provide an additional 60% reduction in LDL-C beyond that achieved with statins alone.<sup>[8]</sup> Furthermore, both agents are associated with a 5–10% increase in HDL-C, an approximate 15% decrease in triglycerides and an approximate 25% decrease in lipoprotein(a).<sup>[8]</sup>

The cardiovascular outcome trials (FOURIER for evolocumab and ODYSSEY Outcomes for alirocumab) showed that the addition of PCSK9 inhibitors to statins significantly reduce the risk of major adverse cardiovascular events. Indeed, evolocumab was associated with 27% relative risk reduction of myocardial infarction and 21% relative risk reduction of stroke. Both agents were associated with 15% reduced relative risk of the composite primary end point.<sup>[29][30]</sup> There are no significant adverse reactions associated with the use of PCSK9 inhibitors, except for injection-site reactions that occur in 2–4% of patients.<sup>[29][30][31]</sup>

The FOURIER open labelled extension study confirmed the cardiovascular benefits of evolocumab over a longer period (>8 y of follow-up) as well as the absence of adverse events, even for patients reaching very low LDL-C concentrations (median LDL-C of 0.8 mmol/L).<sup>[32]</sup>

Both alirocumab and evolucumab should be considered in very high-risk individuals (patients with familial hypercholesterolemia or in secondary cardiovascular prevention) when lipid threshold values are exceeded with maximally tolerated lipid-lowering medication.<sup>[1]</sup>

**Inclisiran** is another type of PCSK9 inhibitor. It is a small interfering RNA (siRNA) that reduces hepatic synthesis of PCSK9. In phase III studies (ORION-10 and ORION-11), this molecule was associated with a reduction in LDL-C of approximately 50% when administered subcutaneously every 6 months.<sup>[33]</sup> The cardiovascular outcome study is currently ongoing (ORION-4).<sup>[34]</sup> Inclisiran has been approved by Health Canada, but use will be influenced by reimbursement considerations.

## Resins

The bile acid-sequestering resins **cholestyramine**, **colesevelam** and **colestipol** reduce plasma LDL-C and can slightly increase HDL-C levels. They have a strong safety record. In the prestatin era, cholestyramine was shown to improve cardiovascular outcomes when used alone as demonstrated by the Lipid Research Council - Cardiovascular Primary Prevention Trial (LRC-CPPT).<sup>[35]</sup> In the modern era, no cardiovascular outcome trial combining these agents with statin therapy has been published. Increase in TG levels, GI distress and concomitant drug malabsorption are the main drawbacks to these agents, although colesevelam may be associated with a better GI tolerability. Resins are appropriate lipid-lowering agents in children and in pregnant or breastfeeding patients. They are also appropriate added to maximally tolerated statin therapy with or without ezetimibe in high-risk patients to help reduce LDL-C.<sup>[1]</sup>

## ANGPTL3 Inhibitors

**Evinacumab** is a monoclonal antibody directed against angiopoietin-like protein 3 (ANGPTL3). Inhibition of ANGPTL3 promotes very low density lipoprotein (VLDL) metabolism in the liver and lowers plasma lipid levels. Evinacumab was approved by Health Canada in 2023 and is indicated as adjunctive therapy in homozygous familial hypercholesterolemia. In the ELIPSE HoFH trial, intravenous infusion of evinacumab every 4 weeks in addition to the background lipid-lowering regimen reduced LDL-C by 49% compared to placebo.<sup>[36]</sup> Its use will be influenced by reimbursement considerations.<sup>[37]</sup>

## Icosapent Ethyl

**Icosapent ethyl** consists of purified eicosapentaenoic acid (EPA, an omega-3 fatty acid), which is derived from fish oil. Icosapent ethyl was approved in Canada to reduce the risk of cardiovascular events. This approval was supported by the results of the REDUCE-IT trial, in which this treatment was associated with a significant 25% relative reduction of major adverse cardiovascular events in patients on statin therapy with fasting TG between 1.52 and 5.63 mmol/L and LDL-C between 1.06 to 2.59 mmol/L. This cardiovascular protection was achieved with 2 g of icosapent ethyl twice daily taken orally (total daily dose, 4 g) over a median of 4.9 years.<sup>[38]</sup> Three types of adverse events occurred significantly more frequently in the icosapent ethyl group than in the placebo group: peripheral edema (6.5% vs. 5.0%), atrial fibrillation (5.3% vs. 3.9%) and constipation (5.4% vs. 3.6%).<sup>[38]</sup> The EVAPORATE trial demonstrated that the addition of icosapent ethyl to statin resulted in slowed coronary plaque progression (low-attenuation, total, total noncalcified, fibrofatty and fibrous plaque volume) compared to statin plus placebo.<sup>[39]</sup>

In its 2021 guidelines, the Canadian Cardiovascular Society recommends the use of icosapent ethyl (2 g BID) to lower the risk of CV events in secondary prevention patients (or primary prevention with diabetes and ≥1 CVD risk factors) who have a fasting TG level between 1.5 and 5.6 mmol/L despite treatment with maximally tolerated statin therapy.<sup>[1]</sup>

## Microsomal Triglyceride Transfer Protein Inhibitors

**Lomitapide** is indicated as adjunctive therapy in adult homozygous familial hypercholesterolemia. Due to its potential for significant hepatic toxicity, regular monitoring of hepatic transaminases is recommended. Where available, referral to a specialized lipid clinic is recommended for patients who may require lomitapide therapy.

## Nicotinic Acid (Niacin)

Niacin is a B vitamin that at high doses lowers triglycerides and LDL-C and raises HDL-C. The AIM-HIGH and HPS2-THRIVE studies failed to show a benefit in increasing HDL-C with niacin in patients already treated optimally with a statin.<sup>[40][41]</sup> Niacin is no longer recommended in the management of dyslipidemic patients.

## Nonprescription Agents

Daily consumption of 10 g **psyllium** may result in a 7% reduction in LDL-C.<sup>[1]</sup> In patients with hypertriglyceridemia, **omega-3 fatty acids** 2–4 g daily may reduce triglycerides by 25–30%.<sup>[1]</sup> Patients taking more than 3 g daily of omega-3 fatty acids from supplements should do so only under a physician's care. High intakes could cause excessive bleeding in patients at high risk for bleeds. Supplementation with long chain omega-3 fatty acids is not associated with CV risk reduction.<sup>[1]</sup> The 2021 Canadian Cardiovascular Society guidelines recommend **plant sterols/stanols** supplementation (≥2 g/day) as a dietary component to reduce LDL-C.<sup>[1]</sup> In a small trial, a product containing 2 g phytosterols (Cardiosmile) showed that compared to placebo, 28 days of supplementation reduced LDL-C and triglycerides by 12% and 14% respectively.<sup>[42]</sup>

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## Choices during Pregnancy and Breastfeeding

Due to hormonal changes, plasma lipids increase throughout the duration of pregnancy; total cholesterol concentrations increase 25–50% and total triglycerides by 2- to 4-fold. By midgestation, there is a 50% increase in LDL-C and a 30% rise in HDL-C, followed at term by a slight decrease in HDL-C.<sup>[43]</sup>

Risk factors for hyperlipidemia also apply to pregnancy and include obesity, weight gain, hypothyroidism, gestational and nongestational diabetes, alcohol consumption, medications, and genetic predisposition.<sup>[44]</sup> Very severe forms of hypertriglyceridemia may increase the risk of pancreatitis.<sup>[45]</sup>

## Prepregnancy Considerations

Risk-factor control is recommended throughout the pregnancy. The teratogenicity of lipid-lowering drugs in humans is not well established but appears to be small (if present at all) with statins, and termination of

pregnancy is not warranted should fetal exposure occur. However, in order to reduce the risks as much as possible, it is recommended to discontinue statins, ezetimibe and PCSK9 inhibitors at least 1 month before stopping contraception. Resins can be safely maintained during pregnancy.<sup>[20]</sup>

### Management during Pregnancy

Reassure patients that unless CV risk is very high, interrupting treatment for the duration of the childbearing period has little impact on the overall risk. Lipid-profile monitoring during pregnancy is not required, except for patients affected by familial hypercholesterolemia (FH) or familial chylomicronemia syndrome (FCS).

### Familial hypercholesterolemia

In the context of FH, which is associated with an extremely high cardiovascular risk, monitoring of LDL-C during pregnancy is required and resins can be safely prescribed. In FH patients with a history of cardiovascular disease or for those with homozygous FH, LDL apheresis can be considered.<sup>[20]</sup>

### Familial chylomicronemia syndrome

In order to avoid the risk of pancreatitis, a very low-fat diet (<10% of calories from fat) is required in these patients, even during pregnancy. Close monitoring of TG value is imperative. Direct removal of triglyceride-rich lipoproteins by plasmapheresis or plasma exchange can be considered in pregnant FCS patients, although the benefits are not well documented (mainly as case reports).<sup>[46][47]</sup>

### Management during Postpartum/Breastfeeding

Treatment may be resumed unless breastfeeding is considered. Lipid-lowering drugs are currently not recommended during breastfeeding due to the lack of safety data.

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

- Use a high-intensity statin (atorvastatin or rosuvastatin) in secondary prevention and very high-CV-risk patients.
- In secondary prevention or very high-risk patients, use a high-intensity statin on alternate days in cases of intolerance rather than change to a less-potent statin.<sup>[16]</sup>
- In the case of insufficient statin efficacy (LDL-C above the thresholds, see [Figure 2](#)), if the statin used is at maximum dose change to a more potent statin.
- Combination therapy is indicated for patients that remain above their LDL-C treatment thresholds despite maximally tolerated statin dose and intensity (see [Figure 2](#)).
- Allow 3 months for a stabilized and representative plasma lipid profile after a major medical event. Acute coronary syndromes require immediate statin therapy regardless of the lipid profile.
- Very low LDL-C (<1.30 mmol/L) is acceptable in high-risk patients, as it appears to be safe and provides additional cardiovascular benefit.<sup>[12][13]</sup>
- Refer any patient that presents with severe hypercholesterolemia (LDL-C  $\geq$ 95th percentile for age and sex), severe HTG (TG  $\geq$ 10 mmol/L) or history of HTG-induced acute pancreatitis to a specialized lipid clinic or a genetic dyslipidemias clinic.

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## Algorithms

**Figure 1:** Estimating the 10-Year Risk of Total Cardiovascular Disease (Framingham Heart Study)<sup>[a][5]</sup>

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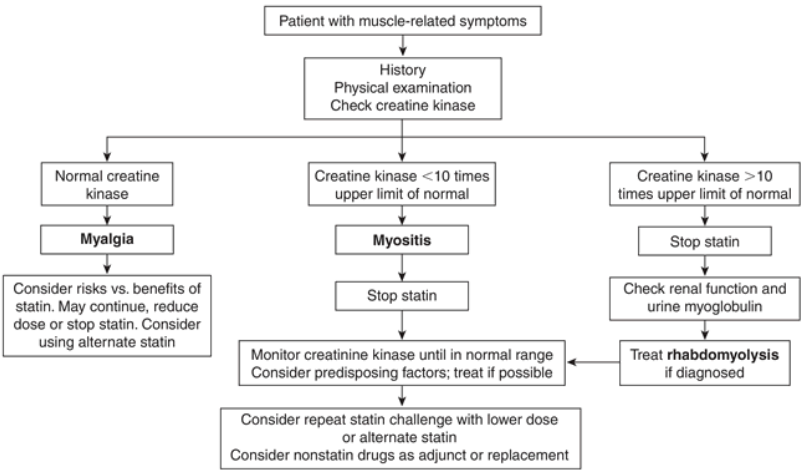
## Therapeutic Tips

- Determine if the dyslipidemia is primary (genetic) or secondary (caused by lifestyle and/or other factors [see [Table 2](#)]).
- Establish the goals of therapy (prevention of cardiovascular disease and/or acute pancreatitis).
- Start with modification of health behaviours (smoking cessation, diet, weight reduction and maintenance, exercise, stress management), preferably with the help of a registered dietitian.
- Adding lipid-lowering drugs must always be an adjunct to, not a substitution for, diet therapy if lipid reduction goals are not achieved.









## Drug Table

Table 7: Drugs Used in the Management of Dyslipidemias

| Drug/Cost <sup>[a]</sup>                                   | Dosage                                 | Adverse Effects                                                                                                                                  | Drug Interactions                                                                                                                                     | Comments                                                                                                                             |
|------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: Angiotensin-converting Enzyme (ACE) Inhibitors |                                        |                                                                                                                                                  |                                                                                                                                                       |                                                                                                                                      |
| <b>lisin</b><br>Ezetrol,<br>generics                       | 10 mg once daily PO at any time of day | Well tolerated.<br>Common: back pain, arthralgia, diarrhea, abdominal pain, fatigue, dizziness, headache.<br><br>Rare: myopathy, rhabdomyolysis, | Low potential for drug interactions. Potential pharmacokinetic interaction with cyclosporine. Caution when initiating ezetimibe in patients receiving | Monitor ALT and CK as indicated.<br>In general, can use in children ≥12 y. Not recommended in moderate to severe hepatic impairment. |

|                                                             |                                    |                                                                           |             |                                                                              |
|-------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------|
| Drug Class: Angiotensin-like Protein 3 (ANGPTL3) Inhibitors |                                    |                                                                           |             |                                                                              |
| <b>evinacumab</b><br>Evkeeza                                | 15 mg/kg IV over 60 min every 4 wk | Dizziness, nausea, infusion reactions (including reports of anaphylaxis). | None known. | Indicated as adjunctive therapy in homozygous familial hypercholesterolemia. |
| \$10,200/345 mg vial                                        |                                    |                                                                           |             |                                                                              |

|                                               |                                        |                                                                                                                                                  |                                                                                                                                                       |                                                                                                                                      |
|-----------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: Cholesterol Absorption Inhibitors |                                        |                                                                                                                                                  |                                                                                                                                                       |                                                                                                                                      |
| <b>ezetimibe</b><br>Ezetrol,<br>generics      | 10 mg once daily PO at any time of day | Well tolerated.<br>Common: back pain, arthralgia, diarrhea, abdominal pain, fatigue, dizziness, headache.<br><br>Rare: myopathy, rhabdomyolysis, | Low potential for drug interactions. Potential pharmacokinetic interaction with cyclosporine. Caution when initiating ezetimibe in patients receiving | Monitor ALT and CK as indicated.<br>In general, can use in children ≥12 y. Not recommended in moderate to severe hepatic impairment. |
| <\$25                                         |                                        |                                                                                                                                                  |                                                                                                                                                       |                                                                                                                                      |

| Drug/Cost <sup>[a]</sup>                                                                                                          | Dosage                                 | Adverse Effects                                                                                                                                                                              | Drug Interactions                                                                                                                                                                                                          | Comments                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: Fibrates                                                                                                              |                                        |                                                                                                                                                                                              |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                         |
| <b>bezafibrate</b> <br>Bezalip SR,<br>generics | 400 mg once daily PO with evening meal | Upper GI disturbances (nausea, abdominal pain, flatulence), myalgias. Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration). | Caution when combining with statins; potential additive hepatotoxicity and myotoxicity. Monitor <a href="#">INR</a> with concomitant warfarin use.<br><br>Increased ezetimibe levels. Monitor for signs of cholelithiasis. | Renal function should be checked at 3, 6 and 12 months, then yearly. ALT checked at least once at 3 months. Check CK if myalgia develops. Useful in diabetic dyslipidemias. Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease. |
| \$25–75                                                                                                                           |                                        |                                                                                                                                                                                              |                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                         |

|                                                                                                                    |                              |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fenofibrate</b> <br>generics | 100 mg BID–QID PO with meals | Upper GI disturbances (nausea, abdominal pain, flatulence), myalgias. Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration). | Caution when combining with statins; potential additive hepatotoxicity and myotoxicity. Monitor <a href="#">INR</a> with concomitant warfarin use.<br><br>Increased ezetimibe levels. Monitor for signs of cholelithiasis.<br><br>Monitor renal function with concomitant cyclosporine use. | Renal function should be checked at 3, 6 and 12 months, then yearly. ALT checked at least once at 3 months. Check CK if myalgia develops. Useful in diabetic dyslipidemias. Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease. |
| \$25–75                                                                                                            |                              |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                                                                                                                             | Dosage                                            | Adverse Effects                                                                                                                                                                                 | Drug Interactions                                                                                                                                                                                                                                                                              | Comments                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fenofibrate</b> <br><b>microcoated</b><br>Lipidil Supra, generics<br><br>\$25–75 | 160 mg once daily PO with largest meal of the day | Upper GI disturbances (nausea, abdominal pain, flatulence), myalgias.<br>Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration). | Caution when combining with statins; potential additive hepatotoxicity and myotoxicity.<br>Monitor <a href="#">INR</a> with concomitant warfarin use.<br><br>Increased ezetimibe levels. Monitor for signs of cholelithiasis.<br><br>Monitor renal function with concomitant cyclosporine use. | Renal function should be checked at 3, 6 and 12 months, then yearly.<br>ALT checked at least once at 3 months.<br>Check CK if myalgia develops.<br>Useful in diabetic dyslipidemias.<br><br>Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease. |
| <b>fenofibrate</b> <br><b>micronized</b><br>generics<br><br><\$25                   | 200 mg once daily PO with largest meal of the day | Upper GI disturbances (nausea, abdominal pain, flatulence), myalgias.<br>Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration). | Caution when combining with statins; potential additive hepatotoxicity and myotoxicity.<br>Monitor <a href="#">INR</a> with concomitant warfarin use.<br><br>Increased ezetimibe levels. Monitor for signs of cholelithiasis.<br><br>Monitor renal function with concomitant cyclosporine use. | Renal function should be checked at 3, 6 and 12 months, then yearly.<br>ALT checked at least once at 3 months.<br>Check CK if myalgia develops.<br>Useful in diabetic dyslipidemias.<br><br>Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease. |
| <b>fenofibrate</b> <br><b>nanocrystals</b>                                        | 145 mg once daily PO with or without food         | Upper GI disturbances (nausea,                                                                                                                                                                  | Caution when combining with statins; potential                                                                                                                                                                                                                                                 | Renal function should be checked at 3, 6 and 12 months, then yearly.                                                                                                                                                                                                                    |

| Drug/Cost <sup>[a]</sup>                                                                                                        | Dosage                                                                                         | Adverse Effects                                                                                                                                                                                 | Drug Interactions                                                                                                                                                                                                                               | Comments                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lipidil EZ</b> , generics<br><br><\$25                                                                                       | Recommended starting dose for patients with renal impairment and in the elderly is 48 mg daily | abdominal pain, flatulence), myalgias.<br>Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration).                                | additive hepatotoxicity and myotoxicity.<br>Monitor <a href="#">INR</a> with concomitant warfarin use.<br><br>Increased ezetimibe levels. Monitor for signs of cholelithiasis.<br><br>Monitor renal function with concomitant cyclosporine use. | ALT checked at least once at 3 months.<br>Check CK if myalgia develops.<br>Useful in diabetic dyslipidemias.<br><br>Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease.                                                                         |
| <b>gemfibrozil</b> <br>generics<br><br><\$25 | 300–1200 mg/day PO divided BID 30 min prior to meals                                           | Upper GI disturbances (nausea, abdominal pain, flatulence), myalgias.<br>Increased bile lithogenicity, increased CK, increased creatinine (not representative of renal function deterioration). | Not to be used in combination with statins.<br>May increase repaglinide and rosiglitazone levels; metabolized by CYP2C8.<br><br>Monitor <a href="#">INR</a> with concomitant warfarin use.                                                      | Renal function should be checked at 3, 6 and 12 months, then yearly.<br>ALT checked at least once at 3 months.<br>Check CK if myalgia develops.<br>Useful in diabetic dyslipidemias.<br><br>Contraindications: hepatic impairment, renal dysfunction, pre-existing gallbladder disease. |
| Drug Class: HMG-CoA Reductase Inhibitors                                                                                        |                                                                                                |                                                                                                                                                                                                 |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                           | Dosage                                          | Adverse Effects                                                                                                                                                                                                                                                                         | Drug Interactions                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">atorvastatin</a><br><a href="#">Lipitor</a> , generics | Adults: 10–80 mg daily<br>PO at any time of day | Common:<br>increased CK,<br>increased<br>transaminases<br>(reversible), mild<br>upper GI<br>disturbances,<br>myalgias (with and<br>without CK<br>elevation), sleep<br>disturbances,<br>headache, rash.<br>Rare: myopathy,<br>rhabdomyolysis,<br>peripheral<br>neuropathy,<br>impotence. | Avoid with CYP3A4<br>inhibitors:<br>amiodarone, azoles,<br>cyclosporine,<br>diltiazem, gemfibrozil,<br>grapefruit juice,<br>macrolide antibiotics,<br>protease inhibitors,<br>verapamil. | Start with low doses<br>and titrate up to reach<br>target doses while<br>monitoring biochemical<br>markers.<br>High-CV-risk patients<br>may start at a high<br>dose.<br><a href="#">ALT</a> should be checked<br>at least once<br>at 3 months. Check CK<br>if myalgia develops.<br>Monitor more closely<br>for adverse effects in<br>patients with moderate<br>to severe renal<br>impairment<br>(<60 mL/min).<br>Contraindications:<br>active liver disease,<br>high alcohol<br>consumption,<br>pregnancy. |

| Drug/Cost <sup>[a]</sup>                | Dosage                                                                                                                                    | Adverse Effects                                                                                                                                                                                                                                                                         | Drug Interactions                                                                              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">fluvastatin</a><br>generics | Adults and Children:<br><a href="#">[b]</a> 20–80 mg PO daily<br>in 1 or 2 divided doses<br><br>Give XL formulation<br>once daily anytime | Common:<br>increased CK,<br>increased<br>transaminases<br>(reversible), mild<br>upper GI<br>disturbances,<br>myalgias (with and<br>without CK<br>elevation), sleep<br>disturbances,<br>headache, rash.<br>Rare: myopathy,<br>rhabdomyolysis,<br>peripheral<br>neuropathy,<br>impotence. | Avoid with CYP2C9<br>inhibitors:<br>amiodarone,<br>fluconazole,<br>fluoxetine,<br>fluvoxamine. | Start with low doses<br>and titrate up to reach<br>target doses while<br>monitoring biochemical<br>markers.<br>High-CV-risk patients<br>may start at a high<br>dose.<br><a href="#">ALT</a> should be checked<br>at least once<br>at 3 months. Check CK<br>if myalgia develops.<br>Monitor more closely<br>for adverse effects in<br>patients with moderate<br>to severe renal<br>impairment<br>(<60 mL/min).<br>Contraindications:<br>active liver disease,<br>high alcohol<br>consumption,<br>pregnancy. |

| Drug/Cost <a href="#">[a]</a>                         | Dosage                                                                                | Adverse Effects                                                                                                                                                                                                                                                                         | Drug Interactions                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">lovastatin</a><br>generics<br><br>\$25–75 | Adults: 20–80 mg daily<br>PO with evening meal;<br>may be given<br>in 2 divided doses | Common:<br>increased CK,<br>increased<br>transaminases<br>(reversible), mild<br>upper GI<br>disturbances,<br>myalgias (with and<br>without CK<br>elevation), sleep<br>disturbances,<br>headache, rash.<br>Rare: myopathy,<br>rhabdomyolysis,<br>peripheral<br>neuropathy,<br>impotence. | Avoid with CYP3A4<br>inhibitors:<br>amiodarone, azoles,<br>cyclosporine,<br>diltiazem, gemfibrozil,<br>grapefruit juice,<br>macrolide antibiotics,<br>protease inhibitors,<br>verapamil. | Start with low doses<br>and titrate up to reach<br>target doses while<br>monitoring biochemical<br>markers.<br>High-CV-risk patients<br>may start at a high<br>dose.<br><a href="#">ALT</a> should be checked<br>at least once<br>at 3 months. Check CK<br>if myalgia develops.<br>Monitor more closely<br>for adverse effects in<br>patients with moderate<br>to severe renal<br>impairment<br>(<60 mL/min).<br>Contraindications:<br>active liver disease,<br>high alcohol<br>consumption,<br>pregnancy. |

| Drug/Cost <a href="#">[a]</a>                        | Dosage                     | Adverse Effects                                                                                                                                                                                                                                                                         | Drug Interactions                                                                                | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">pravastatin</a><br>generics<br><br><\$25 | Adults: 10–80 mg QHS<br>PO | Common:<br>increased CK,<br>increased<br>transaminases<br>(reversible), mild<br>upper GI<br>disturbances,<br>myalgias (with and<br>without CK<br>elevation), sleep<br>disturbances,<br>headache, rash.<br>Rare: myopathy,<br>rhabdomyolysis,<br>peripheral<br>neuropathy,<br>impotence. | Not metabolized<br>through CYP P450<br>pathway; thus, low<br>potential for drug<br>interactions. | Start with low doses<br>and titrate up to reach<br>target doses while<br>monitoring biochemical<br>markers.<br>High-CV-risk patients<br>may start at a high<br>dose.<br><a href="#">ALT</a> should be checked<br>at least once<br>at 3 months. Check CK<br>if myalgia develops.<br>Monitor more closely<br>for adverse effects in<br>patients with moderate<br>to severe renal<br>impairment<br>(<60 mL/min).<br>Contraindications:<br>active liver disease,<br>high alcohol<br>consumption,<br>pregnancy. |

| Drug/Cost <sup>[a]</sup>                                          | Dosage                                                                                                                                                                                                          | Adverse Effects                                                                                                                                                                                                                               | Drug Interactions                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">rosuvastatin</a><br>Crestor,<br>generics<br><br><\$25 | Adults: 10–40 mg daily PO<br><br>Initial dose 10 mg/day PO except in Asian patients <sup>[c]</sup> and those receiving cyclosporine (initial dose 5 mg/day PO)<br><br>Children: <sup>[b]</sup> 5–10 mg daily PO | Common:<br>increased CK, increased transaminases (reversible), mild upper GI disturbances, myalgias (with and without CK elevation), sleep disturbances, headache, rash.<br>Rare: myopathy, rhabdomyolysis, peripheral neuropathy, impotence. | Avoid with CYP2C9 inhibitors: amiodarone, fluconazole, fluoxetine, fluvoxamine.<br>Increased levels with cyclosporin.<br><br>Reduced levels with concomitant use of magnesium/aluminum hydroxide-containing antacids. | Start with low doses and titrate up to reach target doses while monitoring biochemical markers.<br>High-CV-risk patients may start at a high dose.<br><br><a href="#">ALT</a> should be checked at least once at 3 months. Check CK if myalgia develops.<br>Monitor more closely for adverse effects in patients with moderate to severe renal impairment (<60 mL/min).<br><br>Contraindications: active liver disease, high alcohol consumption, pregnancy.<br><br>Administer antacids 2 h after rosuvastatin. |
| <a href="#">simvastatin</a><br>Zocor, generics<br><br><\$25       | Adults: 10–80 mg PO with evening meal<br><br>Children: <sup>[b]</sup> 10–40 mg PO with evening meal                                                                                                             | Common:<br>increased CK, increased transaminases (reversible), mild upper GI disturbances, myalgias (with and without CK elevation), sleep disturbances, headache, rash.<br>Rare: myopathy, rhabdomyolysis, peripheral                        | Avoid with CYP3A4 inhibitors: amiodarone, azoles, cyclosporine, diltiazem, gemfibrozil, grapefruit juice, macrolide antibiotics, protease inhibitors, verapamil.                                                      | Start with low doses and titrate up to reach target doses while monitoring biochemical markers.<br>High-CV-risk patients may start at a high dose.<br><br><a href="#">ALT</a> should be checked at least once at 3 months. Check CK if myalgia develops.<br>Monitor more closely for adverse effects in                                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                                                           | Dosage                                                                                                                                                                                   | Adverse Effects                                                                                                             | Drug Interactions                                                                                                         | Comments                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                    |                                                                                                                                                                                          | neuropathy, impotence.                                                                                                      |                                                                                                                           | patients with moderate to severe renal impairment (<60 mL/min).<br><br>Contraindications: active liver disease, high alcohol consumption, pregnancy.<br><br>Use of 80 mg dose discouraged due to risk of myopathy. |
| Drug Class: Microsomal Triglyceride Transfer Protein Inhibitors                                    |                                                                                                                                                                                          |                                                                                                                             |                                                                                                                           |                                                                                                                                                                                                                    |
| <b>lomitapide</b><br>Juxtapid<br><br>\$32,000                                                      | Initial: 5 mg daily PO for 2 wk<br><br>May increase to 10 mg PO daily if GI side-effects tolerated<br><br>Dose may be increased further at 4 wk intervals<br><br>Maximum: 60 mg PO daily | GI effects (e.g., diarrhea, nausea, vomiting), chest pain, fatigue.                                                         | Avoid use with moderate or strong CYP 3A4 inhibitors (e.g., ketoconazole). Lomitapide will increase statin plasma levels. | Indicated as adjunctive therapy in homozygous familial hypercholesterolemia. Should be taken along with a fat-soluble vitamin supplement. Regular monitoring of liver tests recommended.                           |
| Drug Class: Omega-3 Fatty Acid Derivatives                                                         |                                                                                                                                                                                          |                                                                                                                             |                                                                                                                           |                                                                                                                                                                                                                    |
| <b>icosapent ethyl</b><br><a href="#">Vascepa</a><br><br>\$315                                     | 2 g BID PO                                                                                                                                                                               | Atrial fibrillation, constipation, peripheral edema. Possible allergic reactions in patients allergic to fish or shellfish. | Increased bleeding risk combined with anticoagulant or antiplatelet agents.                                               | The safety of icosapent ethyl in patients with severe hypertriglyceridemia (TG ≥10 mmol/L) has not been determined (risk for pancreatitis and cardiovascular outcomes).                                            |
| Drug Class: Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9) Inhibitor: Monoclonal Antibodies |                                                                                                                                                                                          |                                                                                                                             |                                                                                                                           |                                                                                                                                                                                                                    |

| Drug/Cost <a href="#">[a]</a>                                                                                           | Dosage                                                                                                                                | Adverse Effects                                                                                                                                                                                                                                                              | Drug Interactions                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>alirocumab</b><br><b>Praluent</b><br><br>\$570                                                                       | Initial: 75 mg SC Q2 wk<br><br>May increase to 150 mg SC Q2 wk if required                                                            | Injection site reactions.                                                                                                                                                                                                                                                    | None known.                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>evolocumab</b><br>Repatha<br><br>\$620                                                                               | 140 mg SC Q2 wk or 420 mg SC monthly                                                                                                  | Injection site reactions.                                                                                                                                                                                                                                                    | None known.                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                             |
| Drug Class: Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9) Inhibitor: Small Interfering Ribonucleic Acid (siRNA) |                                                                                                                                       |                                                                                                                                                                                                                                                                              |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>inclisiran</b><br><b>Leqvio</b><br><br>\$3,000/284 mg dose                                                           | Initial: 284 mg SC<br><br>Repeat dose at 3 months, then every 6 months                                                                | Injection site reactions                                                                                                                                                                                                                                                     | None known                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                             |
| Drug Class: Resins                                                                                                      |                                                                                                                                       |                                                                                                                                                                                                                                                                              |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>cholestyramine</b><br>Odan-<br>Cholestyramine,<br>other generics<br><br>\$75–125                                     | Adults: 4 g BID–TID AC PO. Maximum 24 g/day divided in 3 doses<br><br>Children: 240 mg/kg/day PO. Maximum 16 g/day divided in 3 doses | Common: constipation (>10%), bloating, abdominal fullness, flatulence, increased triglycerides, increased transaminases (reversible).<br>Rare: hyperchloremic acidosis, cholecystitis, cholelithiasis, pancreatitis, malabsorption syndrome, GI bleeding, peptic ulceration. | Administer other drugs 1 h before or 4–6 h after resin to limit possible reduced absorption in the GI tract. | Start with once-daily dosing to improve tolerability.<br>Recommend high-fibre diet and high water intake to minimize constipation.<br>Long-term and high-dose use can prevent the absorption of fat-soluble vitamins and folic acid.<br><br>Can be used to lower cholesterol and LDL-C in pregnancy and in children.<br><br>Contraindications: biliary obstruction, dysbetalipoproteinemia, TG >3.4 mmol/L. |

| Drug/Cost <a href="#">[a]</a>                              | Dosage                                                                                                | Adverse Effects                                                                                                                                                                                                                                                              | Drug Interactions                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            |                                                                                                       |                                                                                                                                                                                                                                                                              |                                                                                                              | Powder may be mixed with juice, soup or applesauce. Add granules to at least 90 mL of fluid.<br>Some formulations contain phenylalanine; caution in patients with phenylketonuria.                                                                                                                                                                                                                          |
| <b>colesevelam</b><br>Lodalis,<br>generics<br><br>\$75–125 | Tablets: 2.5 g/day PO daily in 1 or 2 divided doses.<br><br>Maximum 4.375 g/day                       | Common: constipation (>10%), bloating, abdominal fullness, flatulence, increased triglycerides, increased transaminases (reversible).<br>Rare: hyperchloremic acidosis, cholecystitis, cholelithiasis, pancreatitis, malabsorption syndrome, GI bleeding, peptic ulceration. | Administer other drugs 1 h before or 4–6 h after resin to limit possible reduced absorption in the GI tract. | Start with once-daily dosing to improve tolerability.<br>Recommend high-fibre diet and high water intake to minimize constipation.<br>Long-term and high-dose use can prevent the absorption of fat-soluble vitamins and folic acid.<br><br>Can be used to lower cholesterol and LDL-C in pregnancy and in children.<br><br>Contraindications: biliary obstruction, dysbetalipoproteinemia, TG >3.4 mmol/L. |
| <b>colestipol</b><br>Colestid<br><br>\$25–75               | Granules: 5 g BID–TID AC PO. Maximum 30 g/day<br><br>Tablets: 2 g once daily–BID PO. Maximum 16 g/day | Common: constipation (>10%), bloating, abdominal fullness, flatulence, increased triglycerides, increased                                                                                                                                                                    | Administer other drugs 1 h before or 4–6 h after resin to limit possible reduced absorption in the GI tract. | Start with once-daily dosing to improve tolerability.<br>Recommend high-fibre diet and high water intake to minimize constipation.<br>Long-term and high-dose use can prevent the absorption of fat-                                                                                                                                                                                                        |

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                                                                                                                                                                   | Drug Interactions | Comments                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |        | transaminases (reversible).<br>Rare:<br>hyperchloremic acidosis,<br>cholecystitis,<br>cholelithiasis,<br>pancreatitis,<br>malabsorption syndrome, GI bleeding, peptic ulceration. |                   | soluble vitamins and folic acid.<br><br>Can be used to lower cholesterol and LDL-C in pregnancy and in children.<br><br>Contraindications: biliary obstruction, dysbetalipoproteinemia, TG >3.4 mmol/L.<br><br>Powder may be mixed with juice, soup or applesauce. Add granules to at least 90 mL of fluid.<br><br>Some formulations contain phenylalanine; caution in patients with phenylketonuria. |

<sup>[a]</sup> Cost of 30-day supply unless otherwise specified; includes drug cost only.

<sup>[b]</sup> Children 10–17 years of age (boys and postmenarchal girls).

<sup>[c]</sup> Plasma concentrations are influenced by polymorphisms in the SLCO1B1 and ABCG2 genes. Variants that result in higher blood concentrations are more common in patients of Filipino, Chinese, Japanese, Korean, Vietnamese or South Asian descent.<sup>[48]</sup>

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

**Abbreviations:** ALT = alanine transaminase; CK = creatine kinase; GI = gastrointestinal; INR = international normalized ratio; LDL-C = low-density lipoprotein cholesterol; TG = triglycerides

## Suggested Readings

Mancini GB, Baker S, Bergeron J et al. Diagnosis, prevention, and management of statin adverse effects and intolerance: Canadian Consensus Working Group Update (2016). *Can J Cardiol* 2016;32(Suppl 7):S35–S65.

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